Chapter 1

Related Work

The HCI field of exploring the body as an input surface for possible interactions has been widely researched throughout the past years. Interesting topics include the use of wearable systems, various sensing techniques, multi-touch input and more.

In this chapter, related work is discussed in the context of gestural 3D modeling, skin-based fabrication, and fabrication-aware design. The key ideas and limitations of the respective approaches are summarized at the end.

1.1 On-Body Interaction

Our body is our most confidential and familiar interactive system. The skeleton of a human arm affords a natural degree of freedom offering a wide range of application possibilities [1]. Additionally, interactions with the arm can always be accessible without the need for visual awareness to the target device [2]. The aforementioned points combined with the unique context of using our body as a platform for access to communication and computation has proven as a wide area of exploration [3, 4, 5, 6, 7] and results received considerable attention in the HCI community.

Several papers investigate technical approaches for touch-based interactions. The research scope employs the use of capacitive sensing [8, 9], on-body projection [3, 4, 10], ultrasound transmission [11], magnetic [12] and electro-wave propagation [13]. Important in this context is the work "Multi-Touch Skin: A Thin and Flexible Multi-Touch Sensor for On-Skin Input" by Nittala et al. [5]. The group presents a novel fabrication approach for printing thin-layered mutual-capacitance-based touch sensors and is the first to achieve a high-resolution multi-touch capability (cf. 1.1). Their overlay is able to withstand extreme deformations which can occur on local maxima of curvature - such as the wrists or similar joints. Furthermore, their study outcome scores in robustness, functionality and scalability and flags them as features for future projects. However, creating an input surface is regarded to fit a body of any dimension. Hence, a crucial requirement was the development of an interactive and intuitive design tool that can generate sensor layouts - given any arbitrary 2D shape. The group stated that a limitation of the tool was the lack of an automatized body measuring process and that subsequent work could include a scanning task into the existing pipeline to ease iterative refinement. Our approach is tailored to this demand and provides a computer vision method to design thin-layered sensors.

Figure 1.1: Multi-touch skin. Figure acquired from [5]
a) multi-touch input on the forearm for remote interaction;
b) aesthetic bracelet with multi-touch capabilities;
c) input surface close to the ear for audio context; d) sensors for busy hands

Lo et al. [14] introduce in "Skintillates: Designing and Creating Epidermal Interactions" a wearable technology using conductive silver ink on tattoo paper. The thin-layered sensors naturally wrap around the user's skin which facilitates an expressive and personal display. Application examples include capacitive buttons for mobile device control, body position detection and an LED display that reacts to music. One of its fabrication challenges was to provide users, that do not have the technical knowledge, a sophisticated approach for flexible visual and electrical design. Therefore, the production cost of Skintillates was greatly reduced by integrating commonly available materials and easy-to-afford equipment. Nonetheless, the time consumption of building a specific device would highly vary with the design process of a tattoo pattern. By importing our approach into their existing fabrication pipeline, it would enable a flexible and rapid prototyping process for creating any personalized shape.

Recently, skin-friendly materials have been explored as a mean to replace the vastly expensive equipment mostly found in medical domain to produce sensors. A successor to the approach of Skintilattes is "DuoSkin: Rapidly Prototyping On-Skin

User Interfaces Using Skin-Friendly Materials” by Kao et al [8]. The team aims to make a durable on-skin circuitry available to non-experts in the HCI community by utilizing commodity materials. The device is able to sense touch input, display output and communicate wirelessly with other machines. Results from the study show that the fabrication process was fun and enjoyable and that the ability of customization made it more intimate. Nonetheless, the design process uses paper stencils which limits the users in their customization experience. We share the same vision of providing users access to produce sensors without the need of technical background. Furthermore, we believe that our tool can facilitate a faster and more individualistic approach to determine location and size of printed user interfaces.

1.2 Camera-based Interactions

Researchers have advanced to translate physical context into the virtual modeling setting [15, 16, 17]. This trend aids novices to incorporate a better elucidation of the digital environment into their design [18]. A specific work by Andre D. Wilson [19] investigates the application of depth-sensing cameras to identify touch input on a tabletop. Results illustrate that touch sensing is even possible on complex surfaces and allow more interactions e.g. hover states. One step ahead, Dezfuli et al.[6] study in their paper “PalmRC: Imaginary palm-based Remote Control for Eyes-free Television Interaction” the possibilities of leveraging the palm as an interactive surface for TV remote control. They use an RGB-D camera - which is anchored on top of the television - to track gestural movements and identify distinctive tasks. A considerable benefit is the disregard of a calibration task before sensing a different palm. Any user was able to immediately interact with the tool. However, both methods are restricted in their specific location area i.e. it was not possible to communicate with the system out of the cameras perception range. Our intention is to utilize the strength of the camera setup to design a personalized sensor which the user can carry on the body. This provides the freedom of maximum mobility.

Another important contribution, “ExoSkin: On-Body Fabrication” by Gannon et al. [20] uses a custom built extruder to directly print on a drawn path which is projected onto the arm. The designed geometry can then be exported and traditionally processed with a 3D printer and appropriate materials. Nonetheless, the system uses a motion capture system which is considered costly to maintain. Authors stated that other tracking technologies should be considered with the likes of depth-sensing cameras. Initial experimentation with the Kinect resulted in a lack of accuracy. We anchor on this idea by utilizing the second generation version of Kinect which has a more reliable data stream than its predecessor.

1.3 Design Tools in HCI for Fabrications

The aforementioned scientific contributions in HCI have shown the demand for computational means to improve design iterations and sensor overlays for on-body interaction. Researchers already have included external drawing software into their workflow [8, 14].

A novel approach is to engage the non-technical audience in advanced modeling with sketch-based 2D drawings [21, 22]. A different system even incorporates the user to design and fabricate on the material itself [23]. Based on a similar idea, "Tactum: A Skin-Centric Approach to Digital Design and Fabrication" by Gannon et al. [18] present a system to directly use the skin as the design interface. Tactum allows non-experts to intuitively create precise forms on top of the human's body by capturing different gestures e.g. touch, grab, drag and rub. The implementation relies on a single depth camera that senses the input on the body. Additionally, a monitor is mounted on the desk to visualize a 3D representation of the arm and the drawn artwork on it (cf. 1.2a). Results reveal that both experts and non-experts were enthusiastic about the system and further research could stabilize the feasibility of the project. The simplified representation of the arm was a highlighted limitation (cf. 1.2b). It was suggested to portray the precise geometry of the arm and its texture to convey a more steady bridge between interaction and manipulation. Our work aims to investigate this idea further by digitally reconstructing the finer details of a human body through a depth stream.

(a) An illustrative drawing of the setup: The arm is put on the desk

(b) The design environment: 3D model is live updated

Figure 1.2: Tactum. Figure acquired from [18]

This work depicts an initial exploration for skin-centric design.

1.4 Summary

This chapter presented an overview on the current vision and limitations of on-body interaction in the context of sensor design. In the past years, several researchers have together explored the possibilities of sensing techniques and sensor capabilities. Hence, an upcoming challenge and philosophy is to introduce intuitive fabrication methods to users with non-technical background. These works - including ours - depict an initial investigation on skin-centric design. In the next chapter, ideas and findings from prior scientific research are used to facilitate recommendations for our design tool.

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